

# Администрирование локальных сетей

Лабораторная работа №6

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## Цели и задачи работы

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Настроить межсетевую маршрутизацию VLAN в сети с использованием технологии **Router-on-a-Stick** и изучить передачу пакетов между различными VLAN.

## Выполнение лабораторной работы

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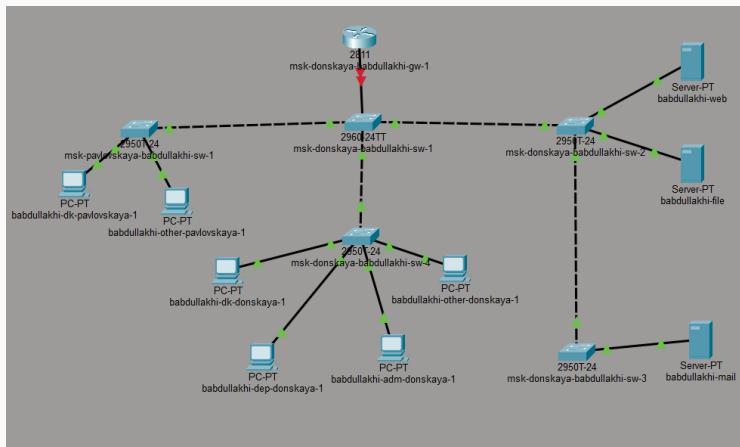


Рис. 1: Топология сети

## Первоначальная настройка маршрутизатора

```
Router>enab
Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname msk-donskaya-babdullakhi-gw-1
msk-donskaya-babdullakhi-gw-1(config)#line vty 0 4
msk-donskaya-babdullakhi-gw-1(config-line)#pass cisco
msk-donskaya-babdullakhi-gw-1(config-line)#login
msk-donskaya-babdullakhi-gw-1(config-line)#line console 0
msk-donskaya-babdullakhi-gw-1(config-line)#pass cisco
msk-donskaya-babdullakhi-gw-1(config-line)#login
msk-donskaya-babdullakhi-gw-1(config-line)#enab secret cisco
msk-donskaya-babdullakhi-gw-1(config)#serv pass
msk-donskaya-babdullakhi-gw-1(config)#user admin privilege 1 secret cisco
msk-donskaya-babdullakhi-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-babdullakhi-gw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-babdullakhi-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-babdullakhi-gw-1(config)#line vty 0 4
*Mar 1 0:2:55.981: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:2:55.981: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-babdullakhi-gw-1(config-line)#transport input ssh
msk-donskaya-babdullakhi-gw-1(config-line)#
```

Рис. 2: Настройка маршрутизатора через CLI

## Настройка VLAN-интерфейсов маршрутизатора

```
msk-donskaya-babdullakhi-gw-1(config-subif)#
msk-donskaya-babdullakhi-gw-1(config-subif)#interface f0/0.101
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.101, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#encapsulation dot1q 101
msk-donskaya-babdullakhi-gw-1(config-subif)#ip address 10.128.3.1 255.255.255.0
msk-donskaya-babdullakhi-gw-1(config-subif)#description dk
msk-donskaya-babdullakhi-gw-1(config-subif)#
msk-donskaya-babdullakhi-gw-1(config-subif)#interface f0/0.102
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.102, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#encapsulation dot1q 102
msk-donskaya-babdullakhi-gw-1(config-subif)#ip address 10.128.4.1 255.255.255.0
msk-donskaya-babdullakhi-gw-1(config-subif)#description departaments
msk-donskaya-babdullakhi-gw-1(config-subif)#
msk-donskaya-babdullakhi-gw-1(config-subif)#interface f0/0.103
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.103, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#encapsulation dot1q 103
msk-donskaya-babdullakhi-gw-1(config-subif)#ip address 10.128.5.1 255.255.255.0
msk-donskaya-babdullakhi-gw-1(config-subif)#description adm
msk-donskaya-babdullakhi-gw-1(config-subif)#
msk-donskaya-babdullakhi-gw-1(config-subif)#interface f0/0.104
msk-donskaya-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.104, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up

msk-donskaya-babdullakhi-gw-1(config-subif)#encapsulation dot1q 104
msk-donskaya-babdullakhi-gw-1(config-subif)#ip address 10.128.6.1 255.255.255.0
msk-donskaya-babdullakhi-gw-1(config-subif)#description other
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

# Проверка связности сети

```
babdullakhi-other-pavlovskaya-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.10

Pinging 10.128.3.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.4.10

Pinging 10.128.4.10 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time<1ms TTL=127
Reply from 10.128.4.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.4.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.5.10

Pinging 10.128.5.10 with 32 bytes of data:

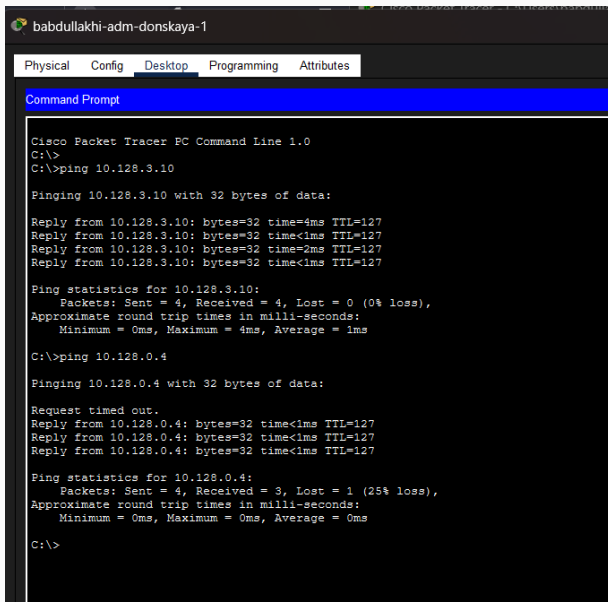
Request timed out.
Reply from 10.128.5.10: bytes=32 time=1ms TTL=127
Reply from 10.128.5.10: bytes=32 time=2ms TTL=127
Reply from 10.128.5.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>
```



## Дополнительная проверка сети



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'babdullakhi-adm-donskaya-1'. The 'Desktop' tab is selected. The Command Prompt displays the results of two ping commands. The first command is 'ping 10.128.3.10', which shows successful results with 0% loss. The second command is 'ping 10.128.0.4', which shows a 25% loss rate.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.128.3.10

Pinging 10.128.3.10 with 32 bytes of data:

Reply from 10.128.3.10: bytes=32 time=4ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127
Reply from 10.128.3.10: bytes=32 time=2ms TTL=127
Reply from 10.128.3.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

# Анализ передачи пакета в режиме Simulation

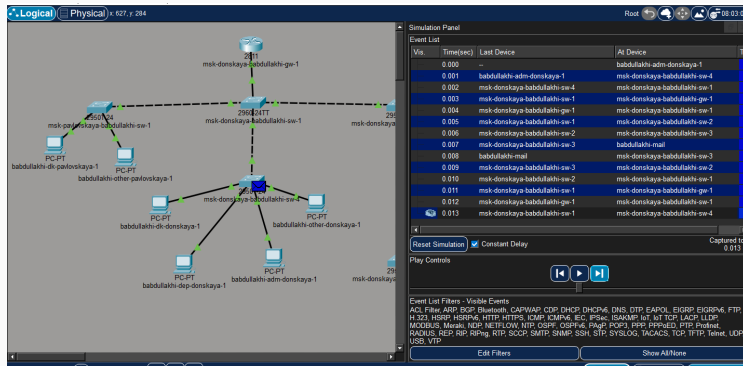
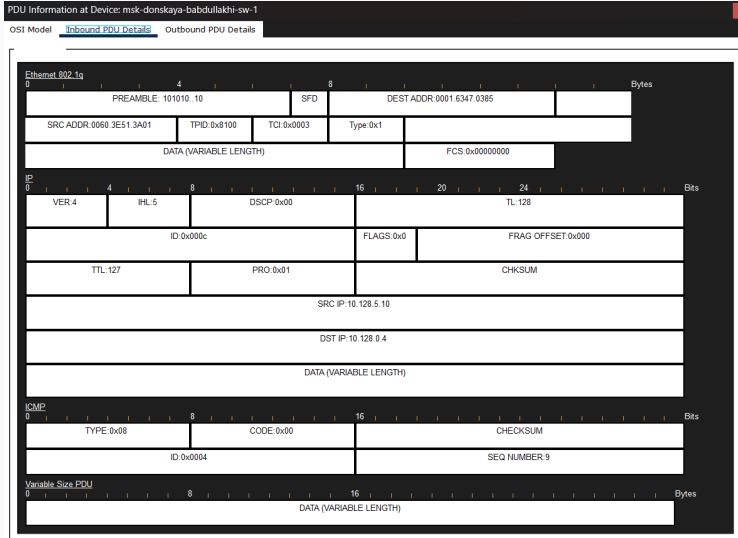


Рис. 6: Перемещение пакета

# Анализ структуры пакета



## Выводы по проделанной работе

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Была выполнена первоначальная настройка маршрутизатора, настроен удалённый доступ по SSH и создана межсетевая маршрутизация между VLAN с использованием подинтерфейсов.

Проверка соединения с помощью команды `ping` показала корректную работу маршрутизации между VLAN. Также был изучен процесс передачи ICMP-пакета и структура сетевых заголовков Ethernet, IP и ICMP.